10	**(a)**	**(i)**	S shown on the peak	B1
		(ii)	Kr and U are right of peak in correct relative positions (Kr on left of U; both on right of S)	B1
		(iii)	Energy released = Binding energy of products – binding energy of reactants $= (144 \times 1.3341 \times 10^{-12} + 90 \times 1.3864 \times 10^{-12}) - (235 \times 1.2191 \times 10^{-12})$	C1
			$= 3.0398 \times 10^{-11}\ J$	A1
		(iv)	$E = mc^2$ $m = \dfrac{3.0398 \times 10^{-11}}{(3.00 \times 10^8)^2}$	C1
			$= 3.38\ (3.3776) \times 10^{-28}$	A1
		(v)	The products have <u>greater stability</u> and <u>therefore greater binding energy.</u> OR With a increase in binding energy, <u>the mass of the products will be less than that of the reactants, by the mass-energy equivalence</u> / <u>mass loss</u>, there must be a release of energy.	B1
		(vi)	Neutrons are single particles, they have <u>no binding energy per nucleon.</u>	B1
	(b)	**(i)**	Isotope is one or more forms of the same element, with the <u>same atomic/proton number but with different number of neutrons in their nuclei / different nucleon number</u>	B1
		(ii)	*Y* = 0 *X* = *-1*	B1
			D = *electron* / beta-particle	B1
		(iii)	Radioactive decay is a random process	M1
			thus the time taken to decay by half will fluctuate. / should consider average time taken.	A1
			Or Carbon-14 will decay into Nitrogen-14	M1
			Wrong, to state that the number of nuclei will decay be half / therefore the total number of nuclei in the box remains the same / Should state “number of carbon-14 nuclei”	A1
	(c)	**(i)**	Since carbon-14 will decay into nitrogen 14, the sample from site will have lower concentration <u>as more time has passed for it.</u> Sample from site has <u>undergone more decay</u> as more time.	B1
		(ii)	Calculating of concentrations or number of nuclei.	C1
			$\lambda = \dfrac{ln2}{5700}$	M1
			$\dfrac{1}{8.6 \times 10^{-10}} = \dfrac{1}{3.3 \times 10^{-10}} e^{-\frac{ln2}{5700}t}$	M1
			$t = 7900\ years\ (7880)$	A1
		(iii)	It cannot be used for very old samples.	B1

			As the activity will be very low after a long period of time and the results of the calculation will not be accurate/reliable.	B1
			It cannot be used for things that are still living	B1
			Carbon-14 could have been gained/ lost via other means.	B1
			Activities from other samples.	B1
			Assumes that the wood will have the same concentration of Carbon-14 to Carbon-12.	B1
			Any two of the above, provided explanations are sound.	